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CANCER INCIDENCE PATTERNS AMONG VIETNAMESE IN THE UNITED STATES AND HA NOI, VIETNAM

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Nearly 600,000 persons have immigrated to the United States from Vietnam since the end of the Vietnam War. Despite the rapid growth of the U.S. Vietnamese population, little is known about cancer incidence in this migrant group. Using population-based data from the Surveillance, Epidemiology and End Results program, California Cancer Registry and International Agency for Research on Cancer, we compared cancer incidence rates for Vietnamese in the United States (1988–1992) to rates for residents of Ha Noi, Vietnam (1991–1993); non-Hispanic whites were included to serve as the U.S. reference rates. Lung and breast cancers were the most common among Vietnamese males and females, respectively, regardless of geographic region. Rates of cancers more common to U.S. whites, such as breast, prostate and colon cancers, were elevated for U.S. Vietnamese compared to residents in Ha Noi but still lower than rates for U.S. whites. Rates of cancers more common to Asian countries, such as stomach, liver, lung and cervical cancers, were likewise elevated for U.S. Vietnamese compared to residents of Ha Noi and exceeded corresponding rates for whites. Incidence patterns for stomach, liver, lung and cervical cancers may reflect increased risk of exposures in this migrant population and should be further explored to uncover the relative contributions of environmental and genetic factors to cancer etiology.

Key words: *Vietnam; migrant; Asian*

The last 20 years have seen an influx of Vietnamese immigrants to the United States as a result of the Vietnam War.[1] By 2000, the U.S. Vietnamese population had grown to over 1.2 million, constituting the fourth largest Asian subgroup in the United States.[2] Despite the rapid growth of this group in recent years, little is known about its cancer incidence patterns. To date, nearly all studies of cancer patterns in U.S. Asian subgroups have been limited to the 3 largest: Chinese, Japanese and Filipino.[3–6] The Vietnamese differ from other Asian migrant groups not only in their more recent immigration but also in their cultural patterns and dietary practices, which potentially relate to differing cancer risks. Two studies have examined cancer incidence patterns in U.S. Vietnamese,[7,8] but they did not compare patterns to those observed in Vietnam to examine the possible effects of migration. As a comparison of cancer incidence and mortality patterns in Vietnamese migrants with patterns in the host population may be useful for better understanding the respective contributions of factors to cancer etiology as well as disparities for cancer control,[9–12] we used the most recent population-based cancer incidence data to examine cancer rates among U.S. Vietnamese and compare them to rates in Ha Noi and U.S. non-Hispanic whites.

MATERIAL AND METHODS

Incidence and population data

For the U.S. populations, we obtained incidence data on invasive cancers from the California Cancer Registry and from the National Cancer Institute's Surveillance, Epidemiology and End Results (SEER) program for the period 1988–1992. For these 2 population-based programs, the catchment area, which includes the states of California, Connecticut, Hawaii, Iowa, New Mexico and Utah and the metropolitan areas of Atlanta, Detroit and Seattle/Puget Sound, covers 52% of the U.S. Vietnamese population. Information on race/ethnicity in these cancer registry data is obtained from the medical record or death certificate. Population estimates by sex, race/ethnicity and age were obtained from the 1990 U.S. Census.

Population-based cancer data for Vietnam were available only from Ha Noi, the capital city located in northern Vietnam. For Ha Noi cases, we obtained incidence data from the International Agency for Research on Cancer for the most comparable time period available, 1991–1993.[13] The Ha Noi Cancer Registry began collecting data on cancer cases regularly in 1988 for the 4 urban districts and 5 rural districts comprising the Greater Ha Noi region. An active search was made for all cancer cases diagnosed in Ha Noi, and information on cancer cases was primarily obtained from the medical record. Cancer reporting is voluntary in the region, and death certificates are not used as a source of information. Although a death registry functions in Vietnam, cause of death information is not required on death certificates. Thus, the completeness of cancer reporting may be affected, particularly for relatively fatal cancers, such as lung and stomach cancers. Because information on race/ethnicity was not available, the statistics reported here for Ha Noi are not delineated by race/ethnicity, though the Vietnamese constitute the largest ethnic group (85–90%) in Vietnam.[14]

To compare the respective public health burden of specific cancers in Vietnam and the United States, we completed the proportional distribution of the 5 most commonly diagnosed cancers for Ha Noi, Vietnamese in the United States and non-Hispanic whites in the United States. For these 3 groups, we also calculated average annual rates for 16 cancer sites, adjusted to the 1970 world standard population[13] using the direct method;[15] 95% confidence intervals were computed based on the γ distribution.[16] Rate ratios were computed to compare age-adjusted cancer incidence rates in U.S. Vietnamese to rates in Ha Noi. We also calculated age-specific rates for major cancer sites (female breast, cervix, prostate, liver, lung, stomach and thyroid) common to Vietnamese in the United States and in Ha Noi.

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RESULTS

Figure 1 shows that, among males, the proportional distribution of the 2 most common cancers, lung and liver, was remarkably similar for Vietnamese in the United States and in Ha Noi, with more than one-third of the overall cancer burden attributed to these 2 cancer types combined. The most striking difference between the 2 geographic areas for Vietnamese was the more common occurrence of prostate and colon cancers (7% and 6%, respectively) in U.S. Vietnamese than in Ha Noi, where these cancers were not among the 5 most frequently diagnosed. Relative to white males, U.S. Vietnamese experienced a smaller proportion of prostate and colon cancers but a much larger proportion of liver and stomach cancers.

Among Vietnamese women, breast cancer was the most commonly diagnosed malignancy in in both the United States and Ha Noi, contributing 18% and 20%, respectively, to the overall cancer burden (Fig. 1). Cervical cancer showed the most marked difference in cancer burden by region; it contributed 5% to the overall cancer burden in Ha Noi but 15% in U.S. Vietnamese. Thyroid cancer was more commonly diagnosed in the United States than in Ha Noi, while stomach cancer was more common in Ha Noi. Although breast and lung cancers were common to both Vietnamese and white females in the United States, cancers of the cervix, thyroid and stomach were among the 5 most commonly diagnosed cancers in Vietnamese, but not in white females.

Tables I and II show age-adjusted incidence rates for major cancer sites for males and females in Ha Noi and the United States. For both sexes, incidence rates for Vietnamese, regardless of region, were generally lower than those of U.S. whites, except for stomach, liver, nasopharyngeal and cervical cancers, while rates were generally higher in U.S. Vietnamese than in residents of Ha Noi, except for nasopharyngeal cancer. Table III compares incidence rate ratios between Vietnamese in the United States and in Ha Noi. Among males, the ratio for prostate cancer incidence was particularly striking, indicating that rates were nearly 35 times higher in the United States than in Ha Noi, the largest discrepancy seen for any cancer site. Among females, cancers related to the gastrointestinal tract (*e.g.,* pancreas, colon and rectum) and reproductive organs (*e.g.,* breast, cervix, corpus uterus and ovary) showed the greatest differences between geographic regions.

Figure 2 shows age-specific incidence rates for the cancers most commonly diagnosed in Vietnamese in the United States and in Ha Noi. Some notable age-specific incidence patterns include that for cervical cancer, for which rates were similar between U.S. whites and Ha Noi residents but exceptionally higher with age in U.S. Vietnamese, with a striking rise at ages 50–64 and rates nearly 7-fold higher than in U.S. white women of the same age group. Breast cancer incidence rates increased substantially after age 64 in white women but peaked in Vietnamese women at ages 50–64, regardless of region and at a much lower level. In Ha Noi and U.S. Vietnamese males, prostate cancer rates were negligible and did not differ across geographic regions until age 65, when rates for U.S. Vietnamese exceeded rates for the same age group in Ha Noi. Rates for liver, stomach and lung cancers increased rapidly with age in U.S. Vietnamese aged 50 and over, diverging markedly from rates for the same age group in Ha Noi. Although thyroid cancer showed similar age-specific patterns in Vietnamese across geographic regions, geographic differences were more pronounced in females than in males.

DISCUSSION

Published information on cancer incidence patterns in Vietnamese in the United States is currently limited, despite the rapid growth of this migrant population. Previous reports of cancer in U.S. Vietnamese populations have been limited by lack of denominators to standardize incidence for intergroup comparisons,[7,8] small sample sizes[8] and lack of international comparisons with

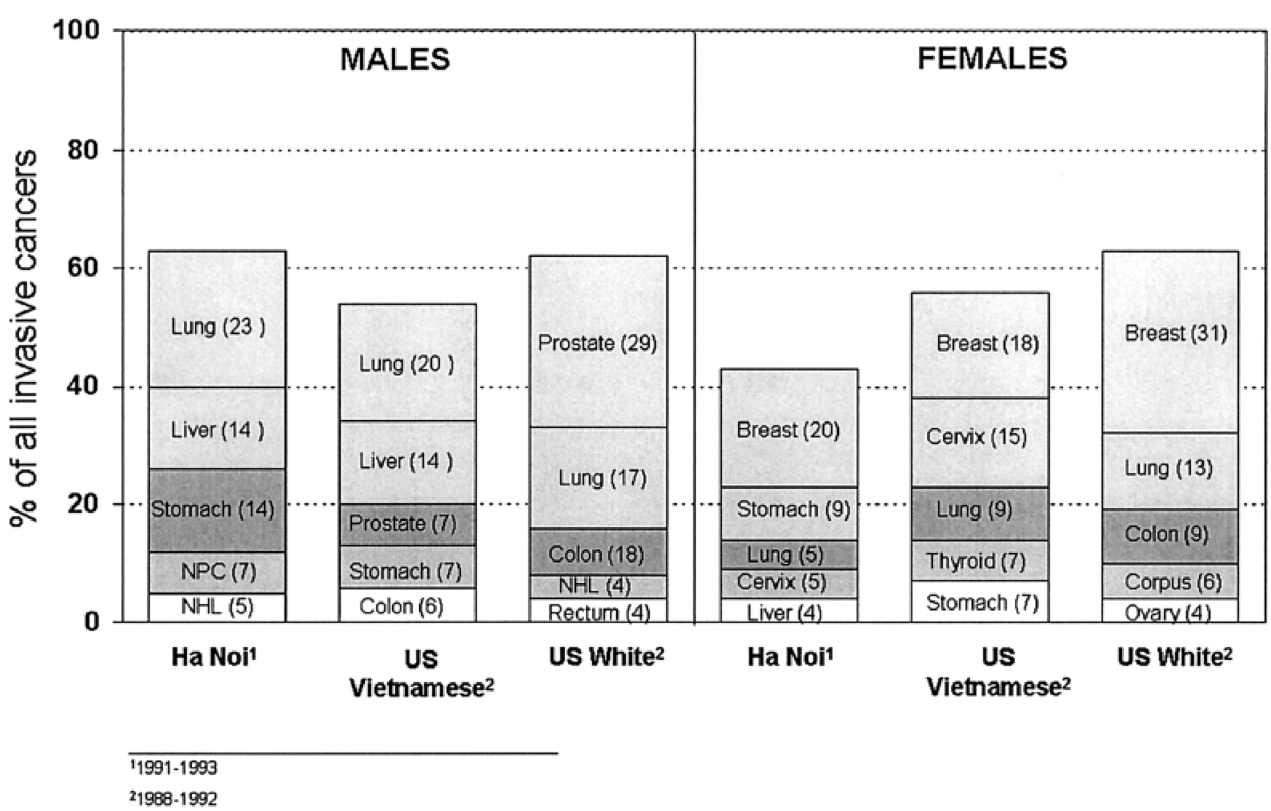

FIGURE 1 – Top 5 cancers contributing most to overall cancer burden.

TABLE I – AVERAGE ANNUAL AGE-ADJUSTED INCIDENCE RATES[1] AND 95% CONFIDENCE INTERVALS (CI) BY CANCER SITE, RACE/ETHNICITY AND REGION FOR MALES

Cancer site	Males								
	Ha Noi[2]			U.S. Vietnamese[3]			U.S. white[3]		
	Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI
Lung	784	29.8	27.7–32.0	210	55.9	48.3–64.0	71,680	64.4	63.9–64.8
Colon	124	4.4	3.6–5.2	64	15.5	11.7–19.8	34,594	29.1	28.7–29.4
Rectum	152	5.6	4.7–6.5	33	7.5	5.0–10.5	15,163	13.4	13.2–13.6
Stomach	475	17.8	16.1–19.5	76	19.8	15.4–24.7	8,347	7.2	7.1–7.4
Liver	487	17.9	16.3–19.6	153	34.6	29.0–40.8	3,089	2.8	2.7–2.9
Pancreas	46	1.8	1.3–2.4	30	7.4	4.9–10.4	9,007	7.9	7.7–8.0
Nasopharynx	250	9.2	8.1–10.4	40	5.9	4.0–8.1	523	0.5	0.5–0.6
Thyroid	30	1.1	0.7–1.5	19	3.2	1.8–5.0	2,559	2.6	2.5–2.7
Bladder	39	1.4	1.0–1.9	27	6.3	4.0–9.1	29,143	24.9	24.6–25.2
Prostate	24	0.7	0.4–1.1	78	24.1	19.0–29.8	123,775	101.0	100.1–101.6

[1]Rates are per 100,000, adjusted to the 1970 world standard population.–[2]Ha Noi, Vietnam (1991–1993).–[3]California + SEER (1988–1992).

TABLE II – AVERAGE ANNUAL AGE-ADJUSTED INCIDENCE RATES[1] AND 95% CONFIDENCE INTERVALS (CI) BY CANCER SITE, RACE/ETHNICITY AND REGION FOR FEMALES

Cancer site	Females								
	Ha Noi[2]			U.S. Vietnamese[3]			U.S. white[3]		
	Count	Rate	95% CI	Count	Rate	95% CI	Count	Rate	95% CI
Lung	172	5.6	4.7–6.5	110	23.8	19.4–28.6	50,546	29.1	28.7–29.4
Colon	80	2.6	2.0–3.2	62	12.7	9.7–16.2	36,467	21.4	21.1–21.6
Rectum	115	3.7	3.0–4.5	41	8.9	6.3–12.0	12,123	8.0	7.8–8.1
Stomach	294	9.5	8.3–10.6	78	15.5	12.1–19.3	4,938	2.9	2.8–3.0
Liver	136	4.5	3.7–5.3	27	5.8	3.7–8.2	1,730	1.2	1.1–1.2
Pancreas	19	0.7	0.4–1.1	28	6.2	4.1–8.8	9,542	5.7	5.6–5.9
Nasopharynx	123	4.1	3.4–4.9	21	3.6	2.1–5.4	248	0.2	0.2–0.3
Thyroid	87	2.8	2.3–3.5	85	12.3	9.6–15.3	6,459	6.4	6.2–6.6
Bladder	6	0.2	0.0–0.3	14	3.2	1.7–5.1	98,921	6.3	6.2–6.5
Breast	500	17.5	16.0–19.1	212	36.6	31.5–42.0	122,871	98.7	98.1–99.3
Cervix	169	5.7	4.8–6.6	178	35.7	30.4–41.5	7,993	7.3	7.2–7.5
Corpus uteri	41	1.5	1.1–2.0	41	8.0	5.7–10.9	23,741	18.5	18.2–18.7
Ovary	58	1.8	1.4–2.4	61	10.6	7.9–13.7	16,964	13.9	13.7–14.2

[1]Rates are per 100,000, adjusted to the 1970 world standard population.–[2]Ha Noi, Vietnam (1991–1993).–[3]California + SEER (1988–1992).

TABLE III – COMPARISON OF RELATIVE CANCER INCIDENCE RATIOS BETWEEN U.S. VIETNAMESE (USV) AND HA NOI (HN)

Male		Female	
Cancer site	USV:HN	Cancer site	USV:HN
Prostate	34.4	Bladder	16.0
Bladder	4.5	Pancreas	8.9
Pancreas	4.1	Cervix	6.3
Colon	3.5	Ovary	5.9
Thyroid	2.9	Corpus uteri	5.3
Non-Hodgkin's lymphoma	2.1	Colon	4.9
		Thyroid	4.4
Liver	1.9	Lung	4.3
Lung	1.9	Non-Hodgkin's lymphoma	3.1
Rectum	1.3		
Stomach	1.1	Rectum	2.4
Nasopharynx	0.6	Breast	2.1
		Stomach	1.6
		Liver	1.3
		Nasopharynx	0.9

which to examine the possible effects of migration on cancer risk.[7,8] To extend the previously published data and contribute more information about cancer in migrant populations, we used the most recent incidence data available to present age-adjusted and age-specific rates for Vietnamese across geographic regions.

The data presented here confirm previous reports of higher incidence of nasopharyngeal, stomach, thyroid (female), cervical and liver cancers and lower incidence of prostate, breast (female), colon and bladder cancers in U.S. Vietnamese compared to the U.S. white population.[7,8] These findings suggest that U.S. Vietnamese had higher incidence of cancers with purported infectious etiologies (*e.g.,* liver, stomach, cervical and nasopharyngeal) but lower incidence of cancers related to hormones (*e.g.,* breast, prostate, ovarian and corpus uteri), westernized diet and lifestyle (*e.g.,* colon). Because 80% of U.S. Vietnamese were foreign-born in 1990,[17] these incidence patterns likely reflect the cancer incidence profile of Vietnamese in their country of origin.

Differences in incidence between U.S. Vietnamese and whites may partly reflect differences in exposure to risk factors. For example, the high rates of liver cancer in Asians have been attributed to their high prevalence of chronic infection with hepatitis B or hepatitis C virus,[18] while high rates of stomach cancer have been linked to diets characterized by high consumption of nitrate-rich and salted foods[19–21] and infection with *Helicobacter pylori*[22] in Asian populations. Infection with human papillomavirus, a well-established risk factor for cervical cancer, is more common in Asian countries and is likely to contribute to the excess of cervical cancer in U.S. Vietnamese compared to white women.[23] Epidemiologic studies of risk factors for nasopharyngeal cancer in Chinese populations have implicated the consumption of salted or fermented foods, particularly childhood consumption of Cantonese-style salted fish, in conjunction with Epstein-Barr virus infection.[24] However, no studies, to our knowledge, have assessed risk factors for these cancers specifically in the Vietnamese population in the United States or Asia. Because the Vietnamese differ from other Asian subgroups in their diet and culture, these studies are worthwhile, to uncover the potential role of cofactors in the development of cancers with established infectious etiologies in this high-risk racial/ethnic population.

Our results also showed marked differences in cancer incidence rates between Vietnamese in the United States and Ha Noi: for

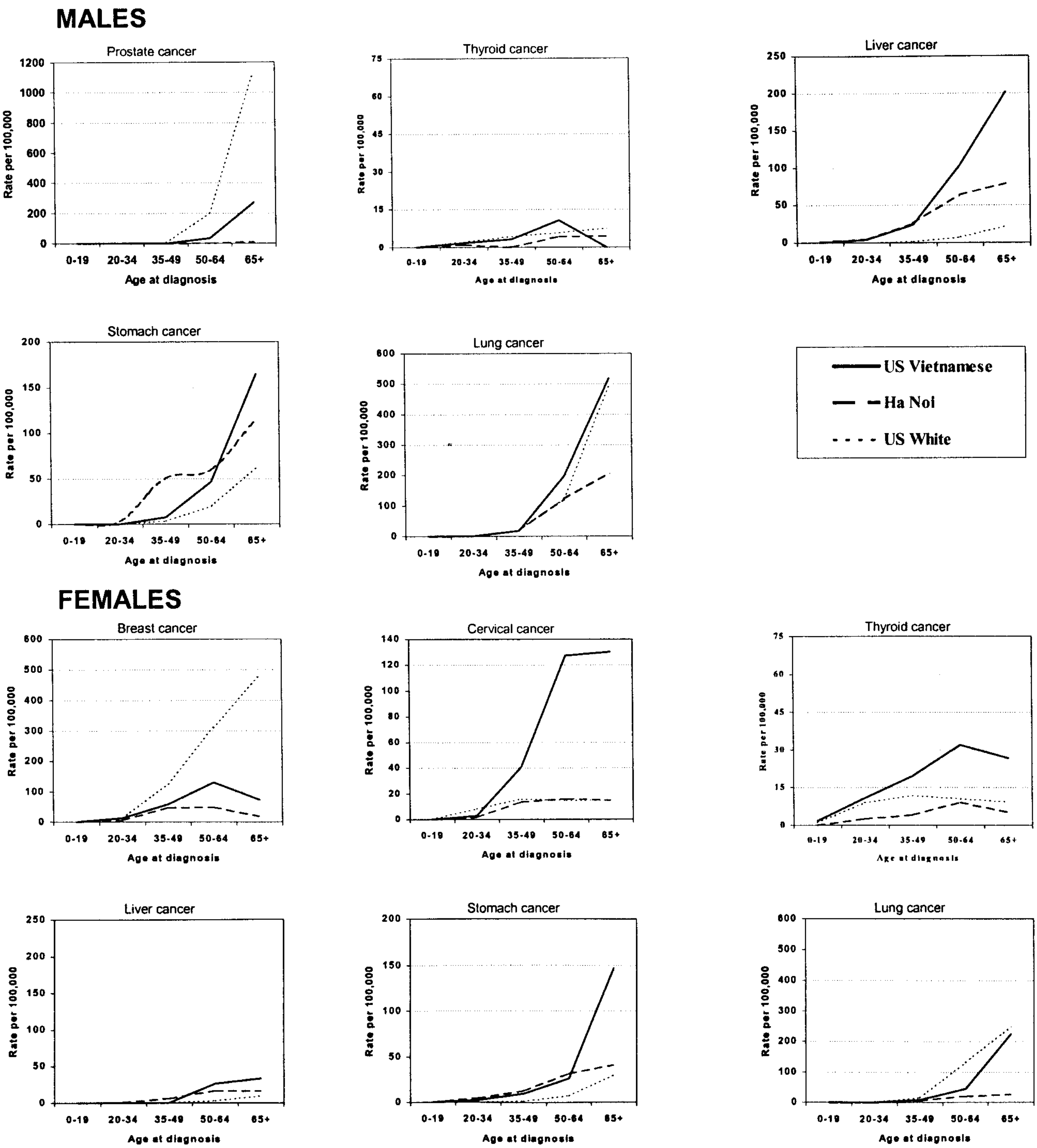

FIGURE 2 – Age-specific incidence rates by gender, region and race/ethnicity.

most cancers, incidence rates were higher in U.S. Vietnamese than in Ha Noi. This general incidence pattern is inconsistent with results from cancer studies in other Asian migrant groups. For Chinese, Japanese and Filipino populations, cancer incidence rates in migrants tend to diverge from rates in their native country and approach those of their host country, suggesting an influence of modifiable risk factors in cancer development.[12,25–27] For example, cancers with an infectious component to etiology, such as liver, stomach and cervical cancers, are uncommon in whites in the United States and tend to occur less frequently in migrants than in their countries of origin.[4,5,12] However, in our data, incidence for these cancers was actually higher in U.S. Vietnamese than in Ha Noi, particularly for liver cancer in males and cervical and stomach cancers in females.

This unexpected pattern raises the question of differences in data quality and comparability across registries, in part due to differences in diagnostic accuracy and case ascertainment.[13] For the U.S. registries included here, the quality of cancer data is relatively high as cancer registration is mandatory and data completeness is estimated to be >97%.[5] However, in Ha Noi, cancer reporting is voluntary and cause of death of information is often missing from death certificates, possibly resulting in underascertainment and, thus, lower rates of relatively fatal cancers, such as lung, liver and stomach cancers. In addition, estimates of the population at risk for computing incidence rates may be high for Vietnamese in the United States as many minority groups were undercounted in the 1990 Census.[28] The quality of the denominator in Ha Noi is likewise subject to uncertainty. These limitations may contribute to the reverse incidence trend seen for these cancer sites.

Rates may also be biased by misclassification of persons by race/ethnicity, particularly in the United States. A study of racial/ethnic misclassification in the San Francisco/Oakland cancer registry showed that 22% of patients who were coded by the registry as Vietnamese did not identify themselves as Vietnamese,[29] though misclassification was more likely to occur in males than in females; after incidence rates were adjusted for misclassification, they were 15% lower in males but only 3% lower in females. To estimate the effect of racial/ethnic misclassification in the United States on incidence patterns, we recalculated our incidence rate ratios by reducing incidence rates in U.S. Vietnamese males and females by 15% and 3%, respectively, and did not find any substantial differences affecting the originally observed patterns (data not shown).

In addition to methodologic influences, the migrant effect, a type of selection bias, should be considered when interpreting differences in incidence rates between migrants and the population of their country of origin.[12] This type of bias occurs because migrants are not typically representative of their population of origin, and the differences may include lifestyle and behavioral characteristics associated with cancer risk. The population covered by the Ha Noi cancer registry may not be comparable to Vietnamese in the United States as these migrants have historically come from southern Vietnam, which differs from the north in culture and diet. We chose Ha Noi rather than Ho Chi Minh City in South Vietnam as the comparison region for its better data quality and comparability standards that have been met for inclusion by the International Agency for Research on Cancer.[13] To determine whether Ha Noi differed from Ho Chi Minh City with regard to the cancer experience, we compared our results with rates published for 1995–1996 from the newly established population-based cancer registry in Ho Chi Minh City.[30] We did not find any substantial differences in cancer incidence between the 2 regions of Vietnam, except for higher rates of cervical cancer (26.0/100,000), lower rates of nasopharyngeal cancer (5.1 and 1.5 in males and females, respectively) and higher rates of liver cancer among males (25.3/100,000) in Ho Chi Minh City than in Ha Noi.[30] Despite regional differences, rates for cervical and liver cancers continue to be higher among U.S. Vietnamese than among residents in Vietnam. Thus, bias due to selective migration from Ho Chi Minh City is not likely to have a substantial impact on the incidence patterns observed here.

Despite methodologic issues of comparability, it can be useful to describe cancer incidence in unique populations for which little information is available as such findings might provide valuable clues to cancer etiology. Studies of cancer in migrants are useful for disentangling the effects of genetic and environmental factors in the development of cancer. If incidence rates among migrants tend to diverge from rates among the population of origin while approaching those of the population of the host country, environmental influences are suggested to play a role in disease etiology. These types of study essentially serve as the initial step in generating hypotheses of etiologic significance. Cancer sites with high incidence rate ratios (Table III) between Vietnamese in the United States and Ha Noi suggest an early influence of environmental exposures, such as those related to lifestyle and diet. The excess incidence of stomach, lung, cervical and liver cancers for Vietnamese in the United States relative to Asia has not been noted in previous migrant studies and warrants further investigation. If replicated, these findings may reflect a cohort effect, possibly due to changes in lifestyle, diet or other exposures as a result of war experiences. It will be fruitful to continue monitoring cancer incidence in this migrant population to observe whether rates of cervical, stomach, lung and liver cancers will decline over time.

Research in migrant populations also may aid in better targeting prevention programs, particularly recent migrants such as the Vietnamese, who may experience low screening rates due to barriers in accessing screening services.[31–34] Our data showed that rates of cervical cancer in U.S. Vietnamese females not only exceeded those of whites but also exhibited a different profile, such that Vietnamese females aged 50–64 had the highest incidence rates, nearly 9-fold higher than rates of whites in the same age group. The high incidence rates in part reflect the low rates of screening in the Vietnamese: results from a 1991 California behavioral risk factor survey showed that 48% of Vietnamese women reported never having a Pap test compared to 5% in the white population.[35] Identifying discrepant patterns of incidence such as those seen here may aid in targeting screening and prevention in this specific population.

Our data highlight the varying rates of cancer incidence among U.S. whites, U.S. Vietnamese and Vietnamese in Ha Noi. Further studies including the Vietnamese population will allow researchers to examine a wider range of exposures associated with cancer risk. This approach would help produce further insights into the relative contributions of heredity and environment to cancer development.

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